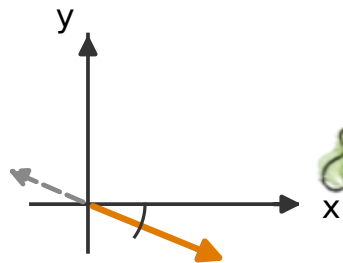




$$\mathbf{s} = (s_x, s_y, s_z), \quad s_z < 0$$

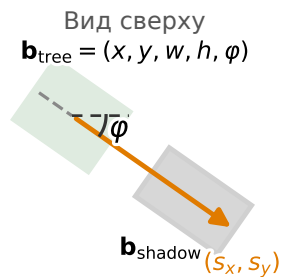
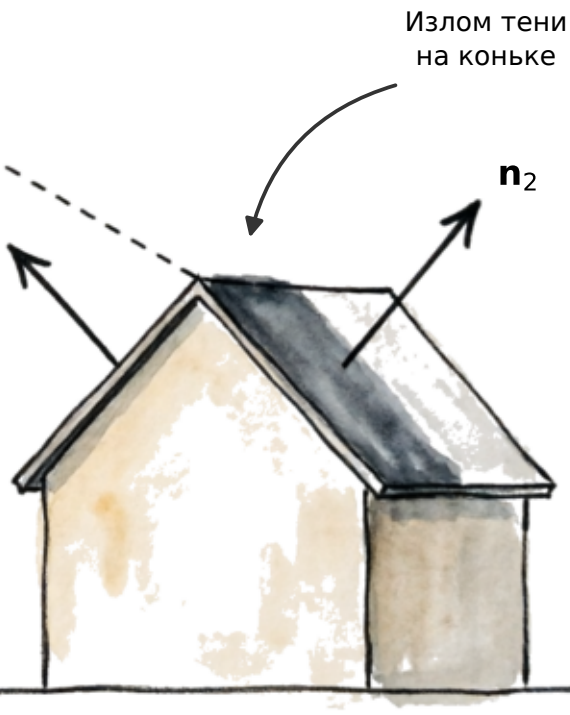
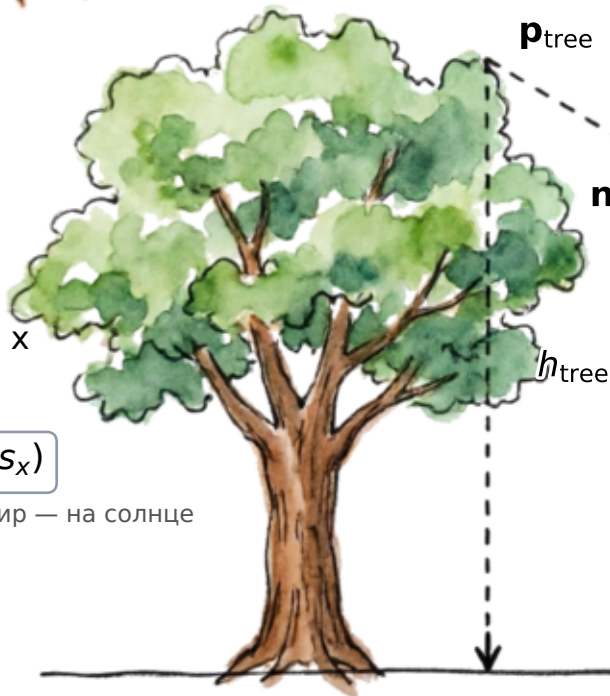
$$\mathbf{p}_{\text{shadow}} = \mathbf{p}_{\text{tree}} - \left(\frac{\mathbf{n} \cdot \mathbf{p}_{\text{tree}} + d}{\mathbf{n} \cdot \mathbf{s}} \right) \mathbf{s}$$

$\mathbf{n}, d$ — нормаль и смещение плоскости ската: $\mathbf{n} \cdot \mathbf{p} + d = 0$



$$\theta_{\text{sun}} = \arctan2(s_y, s_x)$$

(s_x, s_y) — ось тени; пунтир — на солнце



$$\mathcal{F}: I \mapsto \{(\mathbf{b}_i, C_i)\}_{i=1}^K$$

— крона (tree)
— тень (shadow)